Visual Terrain Identification and Surface Inclination Estimation for Improving Human Locomotion with a Lower-Limb Prosthetic

Jean P. Diaz, Rafael L. da Silva, Boxuan Zhong, He Helen Huang and Edgar Lobaton

***Abstract*— Lower-limb robotic prosthetics can benefit from context awareness to provide comfort and safety to the amputee. In this work, we developed a terrain identification and surface inclination estimation system for a prosthetic leg using visual and inertial sensors. We built a dataset from which images with high sharpness are selected using the IMU signal. The images are used for terrain identification and inclination is also computed simultaneously. With such information, the control of a robotized prosthetic leg can be adapted to changes in its surrounding.**

I. Introduction

Humans naturally develop walking capability that is energy efficient, stable, environment adaptive, and robust. Lower limb amputations, unfortunately, disrupt this ability; individuals with lower limb amputations usually depend on prosthetic devices to restore the basic walking function [1].

Recent advances in powered, robotic lower limb prostheses [2], [3], [4] have enabled the efficient performance of locomotor tasks in amputees, such as rising from chairs and climbing stairs and ramps, that are difficult for conventional passive prostheses. However, these robotic devices lack the capacity for environmental adaptation; therefore, a human-machine interface is required so that the user can tell the prosthesis the intended locomotor tasks and the prosthesis can produce appropriate joint control accordingly in the next step [5], [6], [7]. A study designed a body-motion-based human-machine interface to trigger gait initiation, termination, and transitions between sitting and standing [5]. EMG signals recorded from residual muscles were also used as the input of the interface to identify the user intent and enable seamless locomotor task transitions [8]. Fusion of EMG signals and biomechanical sensing were also proposed and tested to recognize the locomotor tasks of the users for prosthesis control [7], [8].

Transition among locomotor tasks (such as level ground walking, stair ascent/descent) requires seamless coordination of amputees and their prostheses according to the environments. In [6], [9], [10], the authors used a combination of a laser distance sensor and an inertial measurement unit (IMU) to recognize the terrain in front of the walking users with different recognition accuracy. The environmental information was modeled as prior probability and integrated into the human-machine interface to improve the accuracy and response time for predicting the users locomotor task transitions. In [11], the authors applied depth sensors to recognize how fast the user approaches a stairs as well as the size of the staircases (e.g. height, depth) . The experimental results demonstrated the potential of visual sensors to enhance the control of lower limb devices. Despite promising results showed in these studies, the applied computer vision method is preliminary and can only provide limited environmental information. In addition, all the tests reported in these study are in well controlled laboratory. It is unclear how the system performs in realistic environments both indoor and outdoor.

In this work, we are focused on propose a terrain recognition system for a prosthetic leg equipped with a camera and one inertial measurement unit (IMU). The camera is placed on the middle of the shin to help in the identification of different terrains. The IMU is placed on the foot to determine when it is nearly static during the gait cycle for capturing stable images, as it will be discussed in the upcoming sections.The rest of the paper is organized as follows: Section II describes the proposed system, Section III shows the results of our analysis, and Section IV provides some final discussions and future work.

II. Proposed System

We envision a sensing platform for a prosthetic leg that incorporates visual as well as inertial sensing in order to identify the type of terrain and its inclination. This information can then be used to provide more context awareness to any control mechanisms. Figure 1 shows the proposed processing pipeline. For our experiments, we mount the sensors on the leg of an able-bodied. A Samsung S6 cellphone is placed at the shin for video collection in daylight conditions, and a Shimmer3 Inertial Measurement Unit (IMU) [12], with acceleration and gyroscope sensors, is placed on top of the foot in a case clipped on a belt strapped around the foot. We use the IMU to determine instances to capture sharp images. Each block of Figure 1 is detailed next.

A. Triggering

As in [13], we have used sharpness as a measurement of image quality. We define sharpness for an image J as:

$$S(J) = \frac{1}{N}\sum_{x,y}\sqrt{\left(\frac{\partial J}{\partial x}(x,y)\right)^2 + \left(\frac{\partial J}{\partial y}(x,y)\right)^2} \quad (1)$$

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where $\frac{\partial J}{\partial x}$ and $\frac{\partial J}{\partial y}$ are the corresponding partial derivatives at pixel $p(x, y)$, and N is the total number of pixels in the image.

The stance phase of a gait cycle takes place from the moment that the foot first makes contact with the ground until the moment when a toe leaves the floor [14]. During this time, the acceleration of the foot is nearly constant, allowing the capture of a low blurred image, J. We call this moment as foot in rest position. Hence, aiming to detect the stance phase for the leg that is carrying the camera, we attached the IMU to the foot of that leg. The position of the IMU on the foot was heuristically determined for that aim, after comparing the signal characteristics by placing the sensor in other parts of the body such as the chest and thigh. As explained next, placing the IMU on the foot provides a signal that enables us to calculate the terrain inclination when the foot is in a rest position.

Our system captures the image J after N_s consecutive IMU samples have a constant acceleration magnitude close to gravity (i.e., when at rest). Hence, we use the notation J_{N_s} to emphasize the dependency on this parameter. In Section III, we analyzed the relation between sharpness and acceleration in order to determine a proper value of N_s.

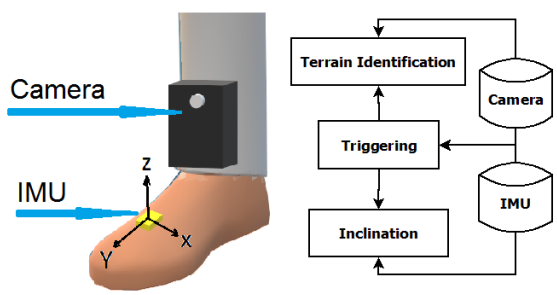

Fig. 1: (Left) Proposed system architecture for visual processing and (Right) Sensors placement on a leg.

B. Inclination

The IMU was used to make the system aware of the current inclination of the ground. In order to reduce the effect of drift due to direct integration of the gyroscope and accelerometer values, several solutions have been proposed [15], [16], [14]. The position where the sensor is placed is crucial to ease signal processing. If placed on the chest for example, the acceleration and gyroscope signals are constantly changing, the same happens by placing it at the thigh. Such constant motion makes it harder to deal with gyroscope drift and consequently, calculating the terrain inclination. Therefore, we placed the IMU on top of the foot as initially proposed by [14], so that the acceleration is nearly constant when the foot is in a rest position. At that instant, our system calculates the angle. This configuration was chosen since there are fewer perturbations (and hence small signal noise) from the dynamics of the leg. The angle is calculated using all axes of the accelerometer and the x-axes of the gyroscope, which is the one that the limb is turning about. The angle of inclination at each sample i is given by:

$$angle(i) = angle_{acc}(i-1) + Gyro_x(i) \cdot \Delta t \qquad (2)$$

where $Gyro_x(i)$ is the gyroscope measurement from its x component at sample i, $\Delta t = \frac{1}{f}$ and $f = 202.84Hz$ is the sampling frequency of the IMU. The term $angle_{acc}(i)$ is given by:

$$angle_{acc}(i) = \arccos\left(\frac{Acc_y(i)}{\|Acc(i)\|^2}\right) \qquad (3)$$

where, $Acc_y(i)$ is the accelerometer measurement from its y component at sample i, $Acc(i)$ is the vector containing the measurements of all components of the accelerometer x,y and z at sample i, and $\|\cdot\|^2$ is the squared Euclidean norm. As one can see, equation (3) only uses accelerometer data, while equation (2) uses both the accelerometer and the gyroscope. The accelerometer is sensitive to noise from the linear movements of the leg. The gyroscope, in the other hand, is not sensitive to linear displacements since it detects angular movements only. The problem with drift that comes when using the gyroscope is avoided since we take measurements when no movements are being executed and such measurements are integrated with the accelerometer data as equation (2) shows.

C. Terrain Identification

The image I_1 is used to identify the terrain type by extracting visual features and classifying the image. Figure 2 shows samples of the different terrain types.

To identify the type of terrain, we used a Bag of Words (BoW) method [17]. The BoW contains the information from features extracted over a regular grid of points 8 pixel apart. Every point in the grid is described using SURF features [18] on every band (RGB) and those three descriptors are concatenated.

III. Experimental Results

We built two datasets, one for testing terrain recognition and another for testing the accuracy of the estimation of inclination. These protocols will be described in Sections III-B and III-C. In both cases, videos and IMU data were collected. Section III-A explores the relationship between image sharpness and acceleration, which was used for triggering of the imaging system.

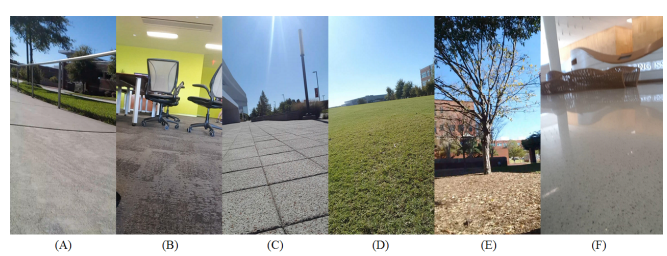

Fig. 2: Samples of the different type of terrains observed from the camera on the leg. (A) Asphalt, (B) Carpet, (C) Cobblestone, (D) Grass, (E) Mulch and (F) Tile.

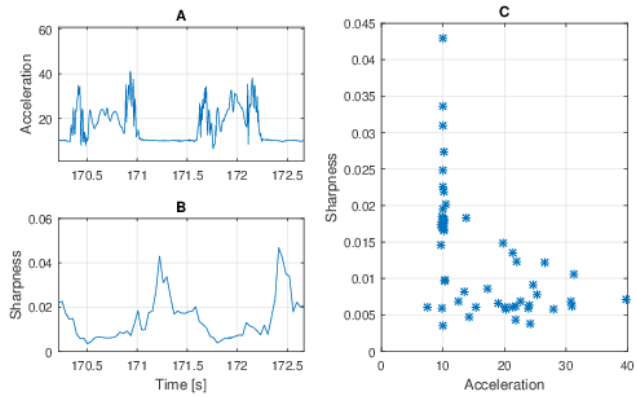

Fig. 3: Relationship between acceleration and sharpness. Waveforms for the (a) magnitude of acceleration and (b) sharpness. (c) Scatter plot shown that images with high sharpness happen at the rest state.

A. Relating Image Sharpness to Acceleration

The protocol to collect data consisted of recording simultaneously videos and IMU data from different subjects, while walking at a regular pace on the North Carolina State University (NCSU) campus for about 20 minutes. The utilized signals come from the accelerometer and gyroscope of the IMU as well as the estimated sharpness levels from the video sequence. The data streams were aligned based on common events indicated by the video and IMU at the beginning and ending of data collection. The subjects jumped in order to cause such events, the acceleration peaks and sound in the video associated with the impact enable us to align both time series. The sound can be used to find the respective video frame within a precision of +/- $\frac{1}{30}s$ These events are manually identified in the data streams and a linear model adjusts the time series of the IMU and video sharpness. Figure 3 shows sample waveforms for the magnitude of acceleration and sharpness. Additionally, the scatter plot shows that the images with the highest values of sharpness correspond to the rest state of the the gait cycle. However, just considering the magnitude of the acceleration at a given time could cause multiple images to be taken at times that may not be appropriate, so we add a wait period N_s before triggering the system. Once N_s samples of the IMU for which the magnitude of acceleration is below τ have been detected, then we capture the image J_{N_s} with corresponding sharpness S_{N_s}.

In order to characterize the quality of S_{N_s} over a single period of the walking gait, we partition the image sharpness data stream into periods starting and ending at consecutive lowest values of sharpness, and take the highest sharpness S_{peak} over that period. We define the quality of our triggered image as $q_{N_s} = S_{N_s}/S_{peak}$. We pick an optimal value of N_s by analyzing the mean quality and the standard deviation of the quality over all the gait cycles in our data. Figure 4 illustrates the range of values of N_s that yield the best performance. From there, we picked $N_s = 62$ since this is the smallest wait periods that gives an acceptable performance.

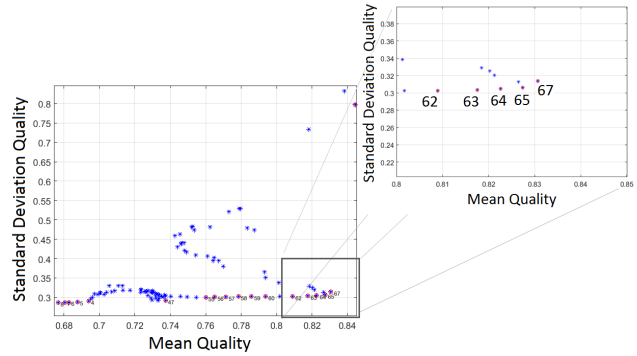

Fig. 4: Pareto Frontier. The highlighted points illustrates that there is a range of values for N_s that yield the lowest standard deviation and highest mean quality.

B. Evaluating Terrain Identification

In order to evaluate the performance of the terrain identification, we built a dataset using the protocol described in Section III-A. In addition to that, the subjects walked over different types of terrains for about 4 minutes each. The dataset consists of 566 images of asphalt, 648 images of carpet, 1041 images of cobblestone, 984 images of grass, 168 images of mulch and 585 of tile (see Figure 2 for samples), which reflect the terrains which the subjects walked over. The images were extracted using the triggering algorithm previously discussed and they were labeled manually. The dataset was randomly divided into a training (80%) and a testing (20%) set. As discussed earlier, classification was done using a BoW approach [17]. Table I shows the Confusion Matrix for the classifier using the test set.

The confusion matrix shows that our classifier is able to identify the 6 types of terrain with an average accuracy of 86%. The grass class yields the lowest performance. It can be mistaken as mulch 26% of the time due to the similar texture on those classes. Also, the tile class can be mistaken as carpet about 18% of the time. With more training data, we expect to be able to train more complex classifiers and hence improve our performance.

C. Evaluating the Inclination Estimation

In order to evaluate the performance of angle estimation, we built a dataset using the protocol described in Section III-A as our base. However, we only collected data on a relatively steep ramp located on NCSU campus. The subject walked from the bottom of the ramp to the top while

TABLE I: Confusion Matrix

	Predicted					
Known	As	Ca	Co	Gr	Mu	Ti
Asphalt	0.88	0.02	0.03	0.02	0.05	0.00
Carpet	0.12	0.71	0.03	0.02	0.08	0.04
Cobblestone	0.07	0.01	0.85	0.01	0.02	0.04
Grass	0.14	0.07	0.05	0.48	0.26	0.00
Mulch	0.02	0.04	0.09	0.03	0.83	0.00
Tile	0.00	0.18	0.01	0.00	0.02	0.79

Fig. 5: Ramp test track with numbered tags. One measurement is taken at each tag and then compared to the calculated angle from the aligned video and IMU streams.

recording the IMU stream and the video from the shin. One extra camera, held by the subject, was also used, pointing to the floor, to record the steps next to the tags for ground truth. The IMU streams and both of the videos are synchronized. In order to obtain a ground truth, we took measurements of the inclination at the mentioned location, 1 meter apart from each other, using a level meter App on a smartphone. Each location was also tagged and numbered (as shown in Figure 5) in order to match them to the video frames taken from the schin and the ones from the camera held by the subject. This allows us to align the measurements and compute an error between them.

Figure 6 shows how close the estimated angle is from the ground truth. We calculated the mean of the absolute error defined as, $MAE = \sum_{k=1}^{N} \frac{1}{N} |x_k - \hat{x}_k|$, where x_k is the ground truth angle at the kth gait cycle and $\hat{x}_k$ is its respective estimated value obtained from Equation (2). The error is averaged over the total number of gait cycles N. Finally, the MAE between the calculated angle and the ground truth is 0.76 degrees. As part of our future work, we will need to determine if this is good enough for integration with control algorithms in the prosthetic leg.

IV. Conclusion

In order to reduce computation of video streams, we proposed a triggering mechanism based on IMU measurements, and optimized for the parameters associated with that mechanism. This triggering is also used to compute terrain inclination at rest states, and consequently avoid the drift errors from the gyroscope and noise from accelerometer. The error obtained by the proposed method for calculating the angle is reasonably low, however, a further assessment of the proposed system could be done by determining the impact of this error on the control modules of a robotized prosthetic solution. The triggering approach enables the capture images with low blurring and consequently provide recognition of various types of terrain. Also, the surface on front of the subject could be estimated to calculate the angle in front of the leg allowing for predicting changes in inclination. As future work, we will aim to validate this approach using more extensive datasets.

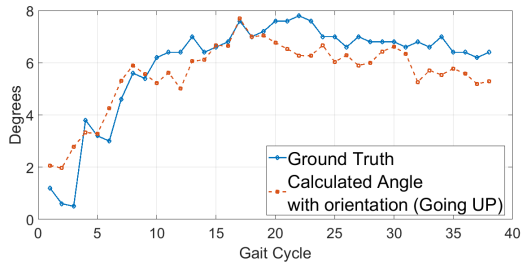

Fig. 6: Calculated and ground truth inclination.

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